

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5

Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Bhuvaneshwaram (192511402) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Resolution Algorithm Implementation

Student 4: Bhuvaneshwaran J | Register Number: 192511402

Question 1 - Machine Auto Shutdown

Knowledge Base

1. If a machine overheats, then the machine shuts down automatically.
2. The machine overheated.

Goal

Prove that **the machine shuts down automatically**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Account Verification

Knowledge Base

1. If a user uploads a valid ID, then the account is verified.
2. The user uploaded a valid ID.

Goal

Prove that **the account is verified**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Premium Subscription Access

Knowledge Base

1. If a customer pays the subscription fee, then the customer gets premium access.
2. Anitha paid the subscription fee.

Goal

Prove that **Anitha gets premium access**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.

Goal

Prove that

4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Smart Device Connectivity

Knowledge Base

1. If a device is connected to WiFi, then the device can access the internet.
2. The smart TV is connected to WiFi.

the smart TV can access the internet. using the Resolution Algorithm.

3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access.** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.

RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	19
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	18
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	13

92

17

Machine Auto Shutdown

let $P \rightarrow$ The machine overheats

$Q \rightarrow$ The machine shut down automatically

$P \rightarrow Q, P$

Goal : Q

CNF:

$\neg P \vee Q$

P

$\neg Q$

Resolve

$(\neg P \vee Q)$ with $(\neg Q)$

Resolvent $\neg P$

Resolve

$(\neg P)$ with P

resolvent (empty clause)

It proves that the system shut down automatically.

Account verification:

let $P \rightarrow$ The user uploads a valid ID

$Q \rightarrow$ The account is verified.

$P \rightarrow Q, P$

Goal : Q

CNF:

$$C_1: \neg P \vee Q$$

$$C_2: P$$

$$C_3: \neg Q$$

Resolve

$$(\neg P \vee Q) \text{ with } (\neg Q)$$

$$\text{Resolvent: } \neg P$$

Resolve

$$(\neg P) \text{ with } P$$

$$\text{Resolvent (Empty clause)}$$

The empty clause is derived, proving that the account is verified.

Premium Subscription Access:

let $P \rightarrow$ Anitha pays the subscription fee

$Q \rightarrow$ Anitha gets premium access

$$P \rightarrow Q, P$$

Goal: Q

CNF:

$$C_1: \neg P \vee Q$$

$$C_2: \neg P$$

$$C_3: \neg Q$$

Resolve $(\neg P \vee Q)$ with $(\neg Q)$

Resolvent

$C_4: \neg P$

Resolve $(\neg P)$ with P

Resolvent

C_5 : Empty Clause

The empty clause is derived, proving that anitha gets premium access.

Smart Device Connectivity:

Let $P \rightarrow$ The Smart TV is connected to wifi

$Q \rightarrow$ The Smart TV can access internet

$P \rightarrow Q, P$

Goal: Q

CNF

$C_1: \neg P \vee Q$

$C_2: P$

$C_3: \neg Q$

Resolve $(\neg P \vee Q)$ with $(\neg Q)$

Resolvent $C_4: \neg P$

Resolve $(\neg P)$ with P

Resolvent C_5 : Empty Clause

The empty clause is derived, proving that the Smart TV can access the internet.

Access Control System:

Let $P \rightarrow$ The user enters the correct password.

$Q \rightarrow$ The user is authenticated

$R \rightarrow$ The user is granted access

$P \rightarrow Q, Q \rightarrow R, P$

Goal: R

CNF:

~~$C_1: P \vee Q$~~

$C_2: \neg Q \vee R$

$C_3: P$

$C_4: \neg R$

Resolve:

$(\neg Q \vee R)$ with $(\neg R)$

Resolvent $C_5: \neg Q$

Resolve

$\neg P \vee Q$ with $(\neg Q)$

Resolvent $C_6 : \neg P$

Resolve

$(\neg P)$ with (P)

Resolvent

C_5 : Empty clause

Since resolving the negated goal with the knowledge base clause leads to an empty clause, a logical contradiction is achieved, successfully proving that the user is granted access.